

CHARLES GATCHALIAN

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Education

University of Houston

Bachelor of Science in Computer Science, GPA: 3.5

Expected December 2027

Houston, TX

Experience

Software Engineering Co-Op

NASA (via LZ Technology, Inc.)

January 2026 – Present

Houston, TX

- Engineered C++ plugin to automate account provisioning in Linux, reducing hypervisor resource overhead by 70%
- Automated RBAC account lifecycles for 20+ Artemis III & IV hosts, translating data into standardized Linux profiles
- Optimized network overhead by 45% by batching calls into LDAP sessions across hypervisors

Undergraduate Research Volunteer

University of Houston

January 2026 – Present

Houston, TX

- Built an LLM-powered agent with long-term memory and function calling to evaluate student code submissions
- Sandboxed student code in Docker containers with memory and CPU limits, blocking malicious process execution
- Accelerated student code evaluation by 40% by parallelizing inference across multiple GPUs using SGLang and PyTorch

Software Engineering Intern

Limbaga Law

May 2025 – August 2025

Houston, TX

- Rebuilt client website in TypeScript with streamlined navigation, reducing support requests by 30%
- Mitigated automated attacks by implementing rate limiting and honeypot filtering, reducing spam submissions by 99%
- Optimized SQL query performance by adding B-tree indexes on read-heavy tables, reducing query latency by 60%

Projects

Pattern Trainer | *Python, JavaScript, PostgreSQL*

- Built a full-stack quiz app helping users identify LeetCode problem patterns across 10 algorithm categories
- Implemented a weighted recommendation algorithm that prioritizes the user's weak categories using PostgreSQL
- Modeled 5 distinct question formats in a JSONB schema to support variable choice and answer structures per type

Task Queueing | *Python, PostgreSQL*

- Built an automated pipeline that generates and categorizes solutions for 75 LeetCode problems using the Gemini API
- Implemented a concurrent job queue with atomic row locking, exponential backoff retry, and a dead letter queue
- Enforced solution quality with Pydantic validation across 4 approach types (optimal, suboptimal, brute force, incorrect)

K8s Portfolio Platform | *Go, JavaScript*

- Deployed a multi-service app on a self-hosted k3s cluster with Helm, TLS termination, and isolated PostgreSQL schemas
- Automated deployment with a CI/CD pipeline using GitHub Actions, running unit tests on every push to the cluster
- Tuned resource limits and Prometheus retention to fit PostgreSQL, Redis, and 3 services within a 4GB VPS budget

Extracurriculars

CougarCS - Information Security Branch

Workshop Officer

August 2023 – May 2025

University of Houston

- Led workshops on Wireshark, packet analysis, and cybersecurity fundamentals for 30+ members
- Increased member turnout by 30% by standardizing beginner materials over Linux setup and setting up cross-org events
- Founded a CTF team and ran weekly collaboration sessions, placing top 18% nationally in PicoCTF

Technical Skills

Languages: TypeScript, Python, C++, JavaScript

Frameworks & Libraries: React, FastAPI, PyTorch, SGLang

Tools & Platforms: Docker, PostgreSQL, GitHub Actions, Linux, AWS, Gemini API

Certifications: CompTIA Security+

Concepts: Object-Oriented Programming, Networking, Data Structures & Algorithms, Agile, RBAC